

Vijay Prajwal Rajasekar

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EDUCATION

University of British Columbia

Expected April 2029

Bachelor of Applied Science – Mechanical Engineering - Aerospace Option

- Co-op Availability: 4, 8, or 12 months

Conestoga College

Graduated

Mechanical Engineering Technology – Automated Manufacturing (Co-op), Cambridge, ON

- Relevant Coursework: Advanced SolidWorks (Sheet Metal, Surface, Weldment), PLC & HMI. Programming, CNC Programming (2D/3D), Additive Manufacturing, Tool & Die and Jigs & Fixture Design.

TECHNICAL SKILLS

CAD & Design Software: AutoCAD, CATIA V5, SolidWorks & 3D-Printing, SolidWorks PDM.

Manufacturing & Programming: PLC-HMI Programming, Robot Controls, PDM/ERP Systems, Time Studies & Downtime Analysis, Tool & Die and Jigs & Fixture Design.

Certifications: Certified SolidWorks Professional (**CSWP**) – 2022; Certified SolidWorks Associate (**CSWA**) – 2020

TECHNICAL WORK EXPERIENCE

Filtovac International Inc. | Breslau, ON

January 2023 - May 2025

Mechanical Engineering Technologist

- Designed 3D CAD models and engineering drawings for individual components and full machine assemblies, streamlining the design-to-manufacturing handoff (SolidWorks)
- Generated Bills of Material for the manufacturing of parts and machines, ensuring accurate procurement and production planning
- Designed models and drawings to support the manufacturing process, improving production efficiency and repeatability
- Prepared as-built drawings and costing reports for completed machines and components, supporting accurate project documentation and client delivery
- Inspected mechanical installations and conducted functional testing of machines, validating build quality against design specifications

Linamar Corporation (Skyjack Inc.) | Guelph, ON

January - April 2021

Manufacturing Engineer Co-op

- Completed the first term of an 8-month Manufacturing Engineer co-op, gaining hands-on exposure to manufacturing engineering processes
- Compiled and analyzed daily assembly reports using Power BI, enabling data-informed decisions across the plant floor
- Conducted process audits and implemented lean manufacturing suggestions, identifying and driving targeted efficiency improvements

Linamar Corporation (Skyjack Inc.) | Guelph, ON

September - December 2021

Manufacturing Engineer Co-op

- Designed a cost-effective, ergonomic weldment jig in SolidWorks, which was manufactured in-house and implemented directly on the production line
- Performed time studies to identify layout improvement opportunities, updating plant layouts and creating detailed standard work instructions
- Delivered continuous improvements across production lines including assembly, paint, and welding areas, increasing overall efficiency

ENGINEERING DESIGN TEAMS

UBCO Aerospace Club | Kelowna, BC

September 2025 - Present

Mech/Manf Sub - Team Lead

- 2026-2027 Project: Working on the AIAA Design/Build/Fly (DBF) Competition.
- Member of UBCO Aerospace Club's Mechanical Team, contributing to the design and manufacturing of aircraft across the club's Fixed-Wing and UAV divisions
- UAV Division: Led the manufacturing of carbon fiber composite components, independently designing mould tooling and developing the complete layup-to-cure fabrication procedure (SolidWorks, composite manufacturing)
- Fixed-Wing Division: Designed structural components for the team's "Guardian" aircraft, including the fuselage and rear V-tail geometry (Onshape CAD)
- Contributed hands-on to the physical build and assembly of the Guardian aircraft, translating CAD designs into a flight-ready structure

TECHNICAL PROJECTS

Involute Gear Train | Personal Project, UBCO Makerspace

January - April 2025

- Modeled a fully parametric, equation-driven involute gear tooth profile in SolidWorks, deriving pitch, base, addendum, and dedendum circles directly from gear theory rather than using standard library gears
- Designed a 4-gear compound train (12T-30T-14T-28T) achieving an overall 1:5 reduction ratio, cross-validating pitch diameters and centre distances against SolidWorks Gear Generator
- 3D printed all four gears in PLA on a Prusa i3 MK3S, iterating on tooth clearance tolerances to resolve binding from the first print and achieve smooth meshing
- Produced a fully functional, hand-driven gear train demonstrating compound gear ratios and speed reduction end-to-end

PLC & HMI Programming Project | Conestoga College (Individual Project)

Coursework Project

- Programmed PLC ladder logic using rungs and various functions, including mathematical operations and loop structures
- Combined timer functions with mathematical operations to control process timing and calculations
- Used mathematical functions to calculate part weight, average weight, and stamping pressure
- Added alarm functions and service-due notifications triggered after set time periods

PROFESSIONAL DEVELOPMENT

Technobytes Technology | IIT-M Research Park, Chennai, India

Drone Technology Workshop

- Completed a two-day, hands-on Drone Technology Workshop covering practical demonstrations and applied drone systems training
- Earned a certification of completion for the program

VOLUNTEER EXPERIENCE

Conestoga College | Kitchener, ON

2020

Orientation Volunteer

- Guided groups of 20 first-year students through campus tours as part of new student orientation
- Answered student questions regarding registration, timetables, and parking, supporting a smooth transition to campus life

AWARDS

SolidWorks Sprint 2025 – Best 1st Year Design | University of British Columbia Okanagan

2025

- Awarded jointly with a teammate; presented by Prof. Dr. Ray Taheri